# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**II Semester of M. Sc. (AOC) Examination April-May 2019**

**OC 728 Organic Name Reactions and Reagents**

**Date: 29/04/2019 Day: Monday Time: 10.00 a.m. to 10.30 a.m. Maximum Marks: 20**

## MCQ

**Important Instructions:**

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

**Q-I Select the correct answer from the four choices given below for each of the following questions. [20]**

1 Which of the following reagent selectively gives cis-alkene upon hydrogenation of alkyne?

(a) $H_2$-Pd (b) $H_2$-Pd/$CaCO_3$ (c) $LiAlH_4$ (d) $NaBH_4$

2 Which of the following statements is true for Grignard reaction?

(a) reagent should be kept in anhydrous alcohol.

(b) dry and inert conditions are not required to be maintained to perform the reaction.

(c) reactivity of the reagent depends only on size of R- group in RMgX.

(d) Grignard reagent cannot be prepared from substrate having acidic hydrogens.

3 Robinson annulation is a combination of ______________reactions.

(a) Mannich and Michael (b) Aldol and Mannich

(c) Michael and Cannizzaro (d) Michael and Aldol

4 Which of the following base is NOT employed in Stork enamine reaction?

(a) pyrrolidine (b) morpholine (c) pyridine (d) piperidine

5 Bicyclic compounds with trans fashion can be synthesized by…………

(a) Grignard reagent (b) dithiane (c) boron reagent (d) Gilman reagent

6 Which of the following reagent is used for "syn" hydroxylation?

(a) Prevost reagent (b) Woodward reagent

(c) Peracid (d) None of above

7 Which of the following reagent is useful for selective oxidation of allylic alcohol?

(a) $SeO_2$ (b) ZnO (c) $MnO_2$ (d) HgO

8 9-BBN does not prefer to undergo migration because ________________.

(a) it is a monohydroborating agent.

(b) secondary carbon is involved in migration.

(c) bridge head carbon is involved in migration which disturbs the cage structure.

(d) double bond formation at bridge head position is not favoured in [3.3.1]system.

9 Bamford-Stevens reaction in protic solvent involves __________intermediate.

(a) carbene (b) carbanion

(c) carbocation (d) none of the above

10 β-amino carboxylic acid in the presence of DCC gives__________________.

(a) α,β-unsaturated carbonyl compound (b) β–hydroxy carbonyl compound

(c) β–lactone (d) β–lactum

11 Which of the following statement is **NOT** true for LDA?

(a) The enolate ion formation is irreversible using LDA.

(b) LDA gives O-alkylated product in minor amount.

(c) Stork-enamine is better alternative of LDA.

(d) LDA is a powerful nucleophile.

12 Which of the following is NOT a monohydroborating agent?

(a) thexylborane (b) 9-BBN

(c) catecholborane (d) disiamylborane

13 Which of the following statements are true for Oppenauer oxidation?

(i) intramolecular hydride transfer occurs through six membered cyclic transition state.

(ii) isopropanol is used as a hydride donor.

(iii) method is less preferred for oxidation of 1°-alcohols to aldehydes.

(iv) hydride acceptor (acetone or cyclohexanone) should be continuously distilled out.

(a) i, iii and iv are correct (b) i, ii and iii are correct

(c) ii, iii and iv are correct (d) Only iii is correct

14 Stereoselectivity of Peterson olefination is governed by _____________.

(a) elimination condition for β-hydroxy silane (b) α-alkyl groups present in β-hydroxy silane

(c) β-alkyl groups present in β-hydroxy silane (d) addition of α-silyl carbanion on carbonyl compound

15 1 mole of a compound gives 3 mole of formic acid, 1 mole of formaldehyde and 1 mole of glyoxylic

acid upon oxidative cleavage through H5IO6. How many moles of H5IO6 will be consumed during the reaction?

(a) 1 mole (b) 2 mole (c) 3 mole (d) 4 mole

**16** Which of the following statements are NOT correct for Hydroboration of alkenes?

(i) It is syn addition of B & H

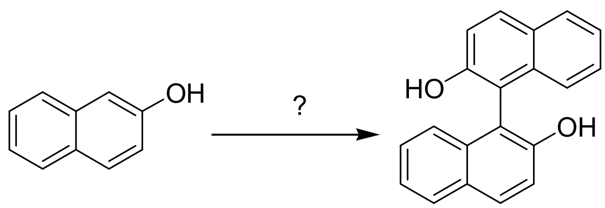
(ii) It is regioselective

(iii) Addition follows anti markovnikov's rule

(iv) Addition favored from less hindered side

(a) i & iii (b) ii, iii & iv (c) iii only (d) iii & iv

**17** The reagent for the following conversion is _______________.




(a) $K_3[Fe(CN)_6]$ (b) n-$Bu_3SnH$

(c) DDQ (d) Alkaline $H_2O_2$

**18** Which of the following products can be synthesized by carbonylation of triethylborane?

(a) 3-ethyl-3-pentanol

(b) acetaldehyde

(c) 2-pentanol

(d) all of the above

**19** Schlosser modification does **NOT** involve ____________________.

(a) ylide formation at -78 °C.

(b) nucleophilic addition of an ylide to a carbonyl compound at room temperature.

(c) proton abstraction from betaine and reprotonation at room temperature.

(d) all of the above steps.

**20** Which of the following reagent is used in the Moffatt oxidation?

(a) $CH_3C_6H_4SO_2Cl$/DMSO (b) DCC/DMSO

(c) DMSO (d) None of above

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**II Semester of M. Sc. (AOC) Examination April-May 2019**

**OC 728 Organic Name Reactions and Reagents**

**Date: 29/04/2019 Day: Monday Time: 10.30 a.m. to 01.00 p.m. Maximum Marks: 50**

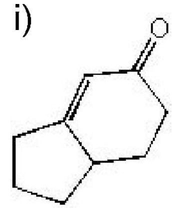
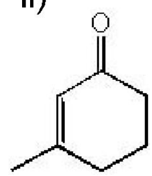
**Important Instructions:**

- **Section I and II must be attempted in TWO ANSWER SHEET.**
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.

## SECTION-I

**Q-II** **Answer the following questions as directed.**

**A** Give the synthesis of following compounds via Robinson Annulation. **04**

i)  ii) 

**B** Give synthesis of a bicyclic compound using thexylborane. **03**

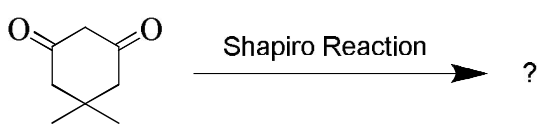
**C** Give synthesis of $Sia_2BH$ and 9-BBN. **03**

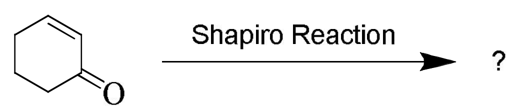
**OR**

**C** Discuss the Schlosser modification in detail. **03**

**Q-III** Answer the following questions as directed.

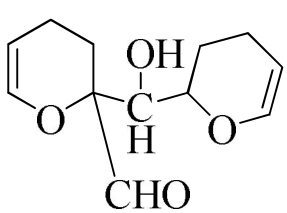
**A** Draw the structure of final product and write the mechanism for it. **04**

i) 


ii) 

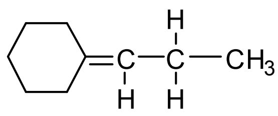

**B** Give synthesis of cyclopentanone using 1,3-dithiane. **03**

**C** Give synthesis of following molecule via Aldol condensation. **03**



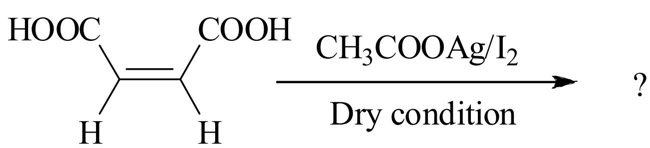

**OR**

| | | |
|---|---|---|
| **C** | Give synthesis of following compound via Wittig reaction. | **03** |

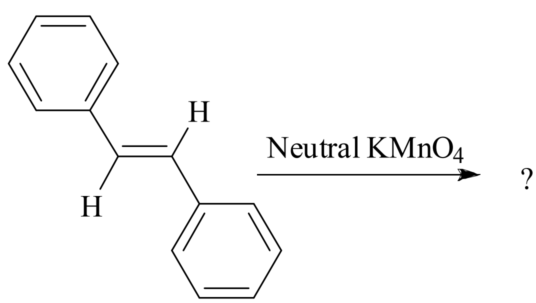


## SECTION-II

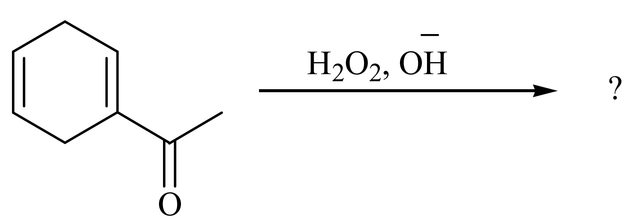
| | | |
|---|---|---|
| **Q-IV** | **Answer the following questions as directed.** | |
| **A** | Discuss: diastereomers will react at different rates with Jone's reagent. | **04** |
| **B** | Give synthesis of cyclohexanol derivative using di-Grignard reagent. | **03** |
| **C** | Discuss Kornblum oxidation with detail mechanism. | **03** |
| | **OR** | |
| **C** | Give synthesis of $\gamma$-lactone using DCC. | **03** |
| **Q-V** | **Draw the structure of correct product and give mechanism.** | |
| **A** | | **04** |



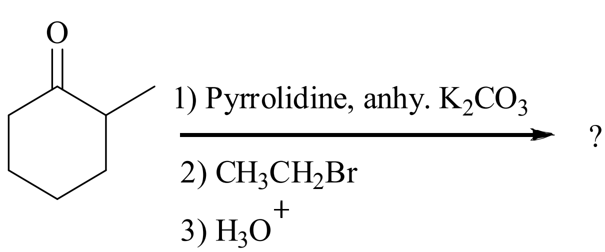

| | | |
|---|---|---|
| **B** | | **03** |



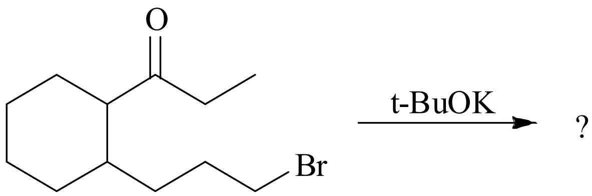

| | | |
|---|---|---|
| **C** | $CH_3MgBr \xrightarrow[\text{2) } H_3O^+]{\text{1) Ethylorthoformate}} ?$ | **03** |
| | **OR** | |
| **C** | | **03** |



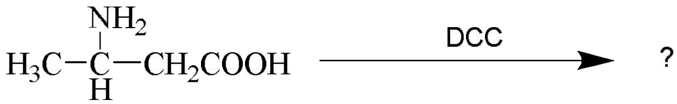

| | | |
|---|---|---|
| **Q-VI** | **Draw the structure of correct product and give mechanism.** | |
| **A** | | **04** |



**B** **03**




**C** **03**




**OR**

**C** **03**